

Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 9: Geometric Twins

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	9 — Geometric Twins
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Watch for SSA dressed as SAS, AAA/AAS misread as congruence, the wrong base angle in algebra problems, conflating line symmetry with rotational symmetry, and assuming equal area or equal perimeter forces congruence. Do not write on this paper — mark answers on your answer sheet.

1. In triangles ABC and PQR, $AB = PQ = 7$ cm, $AC = PR = 7$ cm and $BC = QR = 7$ cm. A student writes "congruent by SAS, since the equal sides force a 60° included angle."

Choose the most precise correct verdict.

- (A) Congruent by SAS, because each triangle is equilateral so the 60° included angle is automatically equal.
- (B) Only similar, because three equal sides fix the shape but the size could still differ.
- (C) Congruence cannot be claimed, since at least one angle measurement is required by every criterion.
- (D) They are congruent, but the criterion is SSS — no angle was given, even though all angles do happen to be 60° .

2. Triangle ABC $\cong$ triangle QRP in that exact letter order. If $\angle A = 35^\circ$ and $\angle B = 85^\circ$, then $\angle P$ equals:

- (A) 35°
- (B) 60°
- (C) 85°
- (D) 120°

3. In triangle ABC, $AB = AC$ and $\angle A = (2x)^\circ$, while each base angle is $(x + 30)^\circ$. The apex angle $\angle A$ measures:

(A) 60°

(B) 30°

(C) 75°

(D) 45°

4. A figure has rotational symmetry of order 4 about its centre but you are asked the smallest positive angle that maps it onto itself, AND how many lines of symmetry this alone guarantees. The correct pair is:

(A) 90° , and exactly 4 lines of symmetry are guaranteed.

(B) 45° , and 4 lines of symmetry are guaranteed.

(C) 90° , and 0 lines of symmetry are guaranteed by order-4 rotation alone.

(D) 120° , and 2 lines of symmetry are guaranteed.

5. Triangles ABC and DEF satisfy $AB = DE$, $\angle A = \angle D$ and $\angle C = \angle F$. A student claims congruence is impossible because the equal angle is not 'between' the side and another angle. The correct verdict is:

(A) Not congruent, because two angles and a side never determine a triangle.

(B) Only similar, since two equal angles fix the shape but leave the size free.

(C) Congruent by SAS, treating the two equal angles as the two equal sides.

(D) Congruent (AAS): with $\angle A = \angle D$ and $\angle C = \angle F$ the third angles match, so $\angle B = \angle E$, and AB lies between A and B, giving ASA.

6. In isosceles triangle ABC, $AB = AC$. The altitude from A meets BC at M, with $AB = AC = 25$ cm and $BC = 14$ cm. The length AM is:

(A) 24 cm

(B) 20 cm

(C) 7 cm

(D) 18 cm

7. Triangle ABC $\cong$ triangle ACB for the SAME triangle (note B and C swapped). Combined with the fact that this self-correspondence maps the triangle onto itself, the strongest forced conclusion is:

(A) $AB = BC$ only, with no angle equality forced.

(B) $\angle A = 90^\circ$, making it a right triangle.

(C) $AB = AC$ and $\angle B = \angle C$, so the triangle is isosceles with apex A.

(D) The triangle must be equilateral.

8. Two triangles have sides $\{5, 12, 13\}$ cm and $\{5, 12, 13\}$ cm respectively. Which statement is the most complete true description?

(A) They are similar but not congruent, because no angle was stated.

(B) They are congruent by SSS and both are right-angled, since $5^2 + 12^2 = 13^2$.

(C) They are congruent by RHS, using the 13 cm hypotenuse and the 5 cm leg.

(D) They cannot be compared without knowing which angle is the right angle.

9. A regular dodecagon (12 sides) is rotated about its centre. The number of distinct positive angles less than 360° (including those that are multiples of the smallest) at which it maps onto itself is:

(A) 12

(B) 11

(C) 6

(D) 1

10. In triangle ABC, AD is drawn so that D is on BC, $BD = DC$, and $\angle ADB = \angle ADC$. Which conclusion is forced, and by which congruence?

(A) $\angle BAC = 90^\circ$, by RHS on the two halves.

(B) Nothing is forced, because two angles at D being equal is automatic for any point on BC.

(C) $AB = AC$: triangles ABD and ACD are congruent by SAS ($BD = DC$, the equal adjacent angles at D, AD common), so the triangle is isosceles.

(D) The triangle is equilateral, by SSS.

11. Triangle PQR has $PQ = PR$ and $\angle P = 20^\circ$. The bisector of $\angle Q$ meets PR at S. Using isosceles base-angle reasoning, $\angle SQR$ equals:

(A) 80°

(B) 20°

(C) 40°

(D) 70°

12. Two triangles ABC and DEF have $AB = DE = 8$ cm, $BC = EF = 6$ cm and $\angle C = \angle F = 50^\circ$. The safest correct verdict about congruence is:

(A) Congruent by SAS, since two sides and an angle are equal.

(B) Not guaranteed — the equal angle (at C, F) is opposite AB/DE, not included between the two equal sides, so this is the ambiguous SSA case.

(C) Congruent by ASA, treating $\angle C$ and the side BC together.

(D) Congruent by RHS, because 50° behaves like a fixed right angle.

13. A figure is mapped onto itself by reflection in a horizontal line and also by reflection in a vertical line through the same centre. The minimum order of rotational symmetry it must have, and the smallest forced rotation angle, are:

(A) Order 2 and 180° .

(B) Order 1 and 360° (no rotational symmetry forced).

(C) Order 4 and 90° .

(D) Order 2 and 90° .

14. In triangle DEF, $DE = DF$ and $\angle D = \angle E + 30^\circ$. Using isosceles symmetry, $\angle F$ equals:

(A) 50°

(B) 80°

(C) 30°

(D) 65°

15. Triangle ABC $\cong$ triangle DEF. It is given that the perimeter is 36 cm, $DE = 13$ cm and $EF = 11$ cm. The length of side CA is:

(A) 13 cm

(B) 11 cm

(C) 12 cm

(D) Cannot be found without an angle.

16. Which capital letter (standard block form) has a horizontal line of symmetry but NO vertical line of symmetry and NO rotational symmetry of order > 1 ?

- (A) H
(C) E
- (B) X
(D) S

17. Two right triangles each have a right angle. The first has legs 7 cm and 24 cm; the second has hypotenuse 25 cm and one leg 7 cm. The correct verdict is:

- (A) Congruent by RHS — the first's hypotenuse is $\sqrt{7^2+24^2} = 25$ cm, so both share hypotenuse 25 cm and a 7 cm leg with the right angle.
(B) Not congruent, since the second triangle's other leg is unknown.
(C) Congruent by SSA, two sides and a non-included angle matching.
(D) Only similar, because one is given by legs and the other by a hypotenuse.

18. The angles of a triangle are in the ratio 3 : 4 : 5. Such a triangle is:

- (A) right-angled, because one ratio part is the largest.
(B) isosceles, because 3 and 4 are close.
(C) scalene and acute (all angles different and each below 90°).
(D) obtuse, because the largest angle exceeds 90° .

19. Triangle ABC has $AB = AC$ and the interior apex angle $\angle A = 140^\circ$. Each base angle of the triangle is:

- (A) 70°
(C) 110°
- (B) 40°
(D) 20°

20. Two triangles ABC and DEF satisfy $\angle A = \angle D = 90^\circ$, $AB = DE = 9$ cm and $BC = EF = 15$ cm (BC and EF are the hypotenuses). The most direct correct criterion is:

- (A) SAS, using the right angle as the included angle between the two given sides.
(B) RHS — equal right angle, equal hypotenuse (15 cm) and an equal leg (9 cm).
(C) SSA, so congruence is not guaranteed.
(D) ASA, since the right angle is one of two equal angles.

21. A regular pentagon and a regular hexagon are drawn. The difference (hexagon minus pentagon) in their number of lines of symmetry, and in their orders of rotational symmetry, are respectively:

- (A) 2 more lines of symmetry and 2 more in rotational order.
(B) 0 (equal lines) and 1 more in rotational order.
(C) 1 more line of symmetry but equal rotational order.
(D) 1 more line of symmetry and 1 more in rotational order.

22. In triangle ABC, the perpendicular bisector of BC passes through A, and separately the perpendicular bisector of AB passes through C. What must the triangle be?

- (A) Equilateral.
(C) Right-angled.
- (B) Isosceles but not equilateral.
(D) Scalene.

23. Triangle ABC $\cong$ triangle PQR. Which single statement of corresponding parts is guaranteed by CPCT?

(A) $\angle B = \angle R$ and $CA = PQ$.

(B) $\angle B = \angle Q$ and $CA = RP$.

(C) $AB = QR$ and $\angle A = \angle Q$.

(D) $\angle C = \angle P$ and $BC = PR$.

24. An isosceles triangle has its apex angle equal to one of its base angles minus 30° . (The two base angles are equal.) The apex angle measures:

(A) 70°

(B) 30°

(C) 55°

(D) 40°

25. A shape is invariant under reflection in exactly one line and under rotation of order 1 (no nontrivial rotation). Which figure fits?

(A) An isosceles (non-equilateral) triangle. (B) A non-square rectangle.

(C) A parallelogram that is neither rectangle nor rhombus.

(D) A regular pentagon.

26. Triangles ABC and DEF have $AB = DE = 10$ cm, $\angle B = \angle E = 35^\circ$ and $BC = EF = 14$ cm. The correct criterion is:

(A) SSA, so congruence is not guaranteed.

(B) SAS — the equal angle ($B = E$) is included between the equal sides AB and BC.

(C) ASA, since one angle and adjacent sides are given.

(D) AAS, treating BC as the angle's opposite.

27. In triangle ABC, $AB = AC = 17$ cm and the altitude from A to BC has length 8 cm. The length of base BC is:

(A) 15 cm

(B) 25 cm

(C) 34 cm

(D) 30 cm

28. Two triangles are reflections of one another across a line. Compared with the original, the reflected triangle has:

(A) a larger area, because two copies of the region now exist.

(B) the same shape but a possibly different size, so it is only similar.

(C) the same area and the same perimeter, but reversed orientation (a left/right flip).

(D) the same orientation, since reflection preserves handedness.

29. Triangle ABC $\cong$ triangle DEF, and in triangle DEF it is known that $\angle D = 2 \cdot \angle E$ and $\angle F = 90^\circ$. The measure of $\angle B$ (in triangle ABC) is:

(A) 60°

(B) 30°

(C) 45°

(D) 90°

30. A square is rotated about its centre. Among the four rotations 90° , 180° , 270° and 360° , the number that map the square onto itself is:

(A) 3

(B) 2

(C) 1

(D) 4

31. In triangle ABC, $AB = AC$ and $\angle A = 100^\circ$. The bisector of base angle B meets AC at D. The measure of $\angle BDC$ is:

(A) 120°

(B) 100°

(C) 60°

(D) 140°

32. Consider: (i) a figure with a line of symmetry must have rotational symmetry of order > 1 ; (ii) a figure with rotational symmetry of order > 1 must have a line of symmetry.

Which is/are FALSE?

(A) Only (i) is false.

(B) Both (i) and (ii) are false.

(C) Only (ii) is false.

(D) Both are true.

33. Triangles ABC and DEF have $\angle A = \angle D$, $\angle B = \angle E$ and the side $BC = EF$. A student says ASA. The most accurate verdict is:

(A) Congruent by ASA, since two angles and a side are given.

(B) Not congruent, because the side is not the included side.

(C) Congruent by AAS — the equal side BC is opposite the equal angle A, not between the two equal angles, so it is AAS rather than ASA, but congruence still holds.

(D) Only similar, since two angles fix only the shape.

34. A figure looks identical after a 60° rotation about its centre. The smallest order of rotational symmetry it can have is:

(A) 60

(B) 6

(C) 3

(D) 1

35. Triangle ABC is isosceles with $AB = AC$. D is the midpoint of BC. A student wants to prove $\angle ADB = 90^\circ$. The cleanest congruence to use on triangles ABD and ACD is:

(A) RHS, assuming $\angle ADB = 90^\circ$ from the start.

(B) SSS — $AB = AC$, $BD = DC$ (D is the midpoint) and AD common — then $\angle ADB = \angle ADC$ by CPCT, and since they are supplementary each is 90° .

(C) ASA, using the two base angles and BC.

(D) SAS, using $\angle BAD = \angle CAD$ which is not yet known.

36. Two triangles satisfy: all three pairs of corresponding angles equal, and one pair of corresponding sides in the ratio 1 : 1. What can be concluded?

(A) Only similar, because AAA never gives congruence even with one equal side.

(B) Congruent by AAA, since all angles match.

(C) They are congruent — equal angles plus one pair of equal corresponding sides fixes the size (AAS/ASA).

(D) Nothing, because angle and side data cannot be combined.

37. In a kite ABCD with $AB = AD$ and $CB = CD$, the diagonal AC is an axis of symmetry. Reflecting the kite across AC, the diagonal BD is mapped:

(A) onto AC, since both diagonals coincide under the reflection.

(B) onto a longer segment outside the kite. (C) onto AB, swapping a diagonal for a side.

(D) onto itself, but with B and D interchanged (BD is perpendicular to AC and bisected by it).

38. Triangle ABC has $\angle A = \angle B$. The line that maps the triangle onto itself by reflection is:

- (A) the line AB itself.
- (B) the perpendicular bisector of BC.
- (C) the perpendicular bisector of AB, equivalently the line through C and the midpoint of AB.
- (D) there is no axis of symmetry.

39. The angles of a triangle are in the ratio 1 : 2 : 3. The triangle is:

- (A) equilateral.
- (B) isosceles and right-angled.
- (C) obtuse and scalene.
- (D) right-angled and scalene.

40. Triangle ABC $\cong$ triangle DEF. Given $\angle A = (3x - 10)^\circ$, $\angle E = (x + 30)^\circ$ and $\angle C = 80^\circ$, with the correspondence $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, the value of x is:

- (A) 20
- (B) 30
- (C) 22
- (D) 25

41. A regular octagon is rotated about its centre. The smallest positive rotation mapping it onto itself, and its order of rotational symmetry, are:

- (A) 45° and order 8.
- (B) 45° and order 4.
- (C) 90° and order 8.
- (D) 22.5° and order 16.

42. Triangle ABC has $AB = AC$. The bisector of $\angle A$ meets BC at D and is extended to meet the circumcircle — but ignoring the circle, what can be said about AD relative to BC, using only congruence?

- (A) AD bisects BC but need not be perpendicular to it.
- (B) AD is the perpendicular bisector of BC: $BD = DC$ and $AD \perp BC$, since triangles ABD and ACD are congruent by SAS ($AB=AC$, $\angle BAD=\angle CAD$, AD common).
- (C) AD is perpendicular to BC but need not bisect it.
- (D) AD is parallel to BC.

43. A triangle and its mirror image across a line are placed side by side. A student claims they cannot be congruent because one is 'left-handed' and one 'right-handed.' The correct evaluation is:

- (A) The student is right; mirror images are only similar, never congruent.
- (B) The student is wrong — a triangle and its reflection are always congruent; congruence permits flips, so reversed handedness does not block it.
- (C) Right for scalene triangles, wrong for isosceles ones.
- (D) Mirror images are congruent only if the triangle is equilateral.

44. In triangle ABC, $\angle A = 90^\circ$ and $AB = AC = 6$ cm. The length of BC is:

- (A) 12 cm
- (B) 6 cm
- (C) $3\sqrt{2}$ cm
- (D) $6\sqrt{2}$ cm

45. Two triangles ABC and PQR have $AB = PQ$, $\angle A = \angle P$ and $AC = PR$. A student writes 'congruent by ASA.' The correct criterion and verdict is:

- (A) ASA is correct, treating the two sides as the two angles.
- (B) SAS — $\angle A$ is included between AB and AC; only one angle is given, so calling it ASA is wrong, but the triangles are congruent.
- (C) SSA, so congruence is not guaranteed.
- (D) AAA, because the remaining angles will match.

46. A figure has rotational symmetry of order 6. The set of all positive angles below 360° that map it onto itself is:

- (A) $\{60^\circ\}$ only.
- (B) $\{60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ, 360^\circ\}$.
- (C) $\{30^\circ, 60^\circ, 90^\circ, 120^\circ, 150^\circ\}$.
- (D) $\{60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ\}$.

47. In triangle ABC, AD is a median to BC and is also the bisector of $\angle A$. The triangle must be:

- (A) isosceles with $AB = AC$.
- (B) right-angled at A.
- (C) equilateral.
- (D) scalene.

48. Triangle ABC $\cong$ triangle DEF with perimeter 45 cm. If $DE : EF : FD = 4 : 5 : 6$, the length of side BC is:

- (A) 15 cm
- (B) 12 cm
- (C) 18 cm
- (D) 9 cm

49. Which English capital letter (block form) has rotational symmetry of order 2 AND at least one line of symmetry?

- (A) S
- (B) N
- (C) I
- (D) P

50. Triangle ABC has $AB = AC = 5$ cm and $BC = 8$ cm. The length of the altitude from A to BC, and the area of the triangle, are:

- (A) 4 cm and 16 cm^2 .
- (B) 3 cm and 24 cm^2 .
- (C) 4 cm and 20 cm^2 .
- (D) 3 cm and 12 cm^2 .

51. A motif is repeated around a centre by rotating it through 40° steps, with no reflection symmetry. Its order of rotational symmetry and number of lines of symmetry are:

- (A) order 9 and 9 lines of symmetry.
- (B) order 40 and 0 lines of symmetry.
- (C) order 9 and 0 lines of symmetry.
- (D) order 8 and 0 lines of symmetry.

52. Triangles ABC and ABD lie on opposite sides of common side AB, with $AC = AD$ and $BC = BD$. Joining C to D, what is the relationship of line AB to segment CD?

- (A) AB is parallel to CD.
- (B) AB passes through C and D.
- (C) AB bisects CD but is not perpendicular to it.
- (D) AB is the perpendicular bisector of CD.

53. In triangle DEF, $DE = DF$ and $\angle E = 2 \cdot \angle D$. The measure of $\angle D$ is:

(A) 72°

(B) 45°

(C) 36°

(D) 30°

54. A figure is mapped onto itself by a 120° rotation and by reflection in three lines through its centre. Which figure exactly fits, and what is its rotational order?

(A) A regular hexagon, rotational order 3. (B) An equilateral triangle, rotational order 3.

(C) An equilateral triangle, rotational order 6. (D) A square, rotational order 3.

55. Triangle $ABC \cong$ triangle BCA for the SAME triangle (a 'cyclic' self-correspondence $A \rightarrow B \rightarrow C \rightarrow A$). What does this force?

(A) $AB = BC$ only, so isosceles with apex B. (B) $\angle A = 90^\circ$.

(C) All three sides equal and all three angles equal — the triangle is equilateral.

(D) Nothing — every triangle satisfies this.

56. Two triangles have equal perimeters of 24 cm, and the first has sides 6, 8, 10 cm. The second is known to be congruent to the first. The second triangle's largest angle is:

(A) 90°

(B) 60°

(C) less than 90° (acute).

(D) cannot be determined.

57. A non-square rhombus and a non-square rectangle are compared. The statement that is TRUE for both is:

(A) Each has 4 lines of symmetry and order-4 rotation, like a square.

(B) Each has 0 lines of symmetry but order-2 rotation.

(C) The rhombus has 2 lines and the rectangle has 0 lines of symmetry.

(D) Each has exactly 2 lines of symmetry and rotational symmetry of order 2.

58. In triangle ABC , $AB = AC$ and a point D lies on BC with $\angle BAD = \angle CAD$. If $\angle ABC = 50^\circ$, then $\angle ADB$ equals:

(A) 90°

(B) 50°

(C) 80°

(D) 65°

59. Two figures are congruent. One is obtained from the other by a single rotation (no reflection). Comparing the two, which statement is fully correct?

(A) They have equal corresponding sides and angles, equal area and perimeter, and the SAME orientation (handedness preserved).

(B) Equal sides and angles, but reversed orientation, since any rigid motion flips handedness.

(C) Equal area but possibly different perimeter, since rotation can stretch one direction.

(D) Only similar, because a rotation can change the size.

60. Point P is reflected across line m to give P' , and the segment PP' has length 10 cm. Line m meets PP' at point F . Which statement about F is always correct?

(A) F divides PP' so that $PF = 10$ cm and $FP' = 0$ cm.

(B) m is parallel to PP' , so F does not exist.

(C) $PF = 7$ cm and $FP' = 3$ cm, depending on the tilt of m .

(D) F is the midpoint of PP' and $m \perp PP'$ there, so $PF = FP' = 5$ cm.

Solutions — Mathematics (Class 7), Chapter 9: Geometric Twins — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	D	31	A	41	A	51	C
2	B	12	B	22	A	32	B	42	B	52	D
3	A	13	A	23	B	33	C	43	B	53	C
4	C	14	A	24	D	34	B	44	D	54	B
5	D	15	C	25	A	35	B	45	B	55	C
6	A	16	C	26	B	36	C	46	D	56	A
7	C	17	A	27	D	37	D	47	A	57	D
8	B	18	C	28	C	38	C	48	A	58	A
9	B	19	D	29	B	39	D	49	C	59	A
10	C	20	B	30	D	40	A	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) All three corresponding sides are equal, which is exactly the SSS criterion; SSS needs no angle. Naming it SAS misstates which given facts were used — no angle was supplied as data, so the criterion invoked must be SSS. The triangles ARE congruent (so 'only similar' and 'cannot claim' are wrong), but the justification is three equal sides, not an included angle.

2. (B) From the order $ABC \leftrightarrow QRP$, vertex C corresponds to P, so $\angle P = \angle C$. In triangle ABC, $\angle C = 180^\circ - 35^\circ - 85^\circ = 60^\circ$. Reading P as matching A (35°) or B (85°) ignores the third-letter correspondence.

3. (A) Angle sum: $2x + (x+30) + (x+30) = 180$, so $4x + 60 = 180$, giving $4x = 120$, $x = 30$. The apex $\angle A = 2x = 60^\circ$. Reporting x itself (30) as the apex, or reading the base angle $x+30 = 60$ wrongly, gives the distractors.

4. (C) Smallest rotation = $360^\circ/4 = 90^\circ$. Rotational symmetry of order 4 does not by itself force any reflection axis (e.g. a 'pinwheel' shape turns onto itself every 90° yet has no

mirror line), so 0 lines are guaranteed. Choosing 45° halves the correct step; assuming 4 mirror lines confuses rotational order with reflection count.

5. (D) Two pairs of equal angles force the third pair equal (angle sum), so $\angle B = \angle E$. Now AB is the side between the equal angles A and B in one triangle and DE between D and E in the other — that is ASA (the AAS data reduces to ASA). Two-angles-plus-a-side DOES fix a triangle, so 'not congruent' and 'only similar' are false.

6. (A) The altitude from the apex of an isosceles triangle bisects the base, so $BM = MC = 7$ cm. In right triangle ABM, $AM = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ cm. Forgetting to halve the base (using 14) or mis-subtracting gives the distractors.

7. (C) The correspondence $A \leftrightarrow A, B \leftrightarrow C, C \leftrightarrow B$ equates AB with AC and $\angle B$ with $\angle C$ (CPCT). That makes the triangle isosceles with apex A. It need not be equilateral (the third side BC is free) and $\angle A$ is unconstrained, so the right-angle and equilateral options overreach.

8. (B) Equal triples of sides give SSS congruence directly. Separately, $25 + 144 = 169 = 13^2$, so each triangle is right-angled — that is an extra true fact, not a missing requirement. RHS is the wrong NAME of the criterion here (we used three sides, not the right-angle/hypotenuse/leg package), and equal sides give congruence, not mere similarity.

9. (B) A regular 12-gon maps onto itself at every multiple of $30^\circ (= 360^\circ/12)$. The positive angles below 360° are $30^\circ, 60^\circ, \dots, 330^\circ$ — that is 11 distinct rotations. The $360^\circ (= 0^\circ)$ return is the identity and is not a positive angle below 360° , so 12 overcounts.

10. (C) Since $\angle ADB$ and $\angle ADC$ are equal and sit on the straight line BC, each is 90° , and $BD = DC$ with AD common. Triangles ABD and ACD then match by SAS (two sides and the included right angle), so $AB = AC$ by CPCT — isosceles. The angles at D are NOT automatically equal for a general D, and nothing forces $\angle A = 90^\circ$ or all sides equal.

11. (C) Base angles: $\angle Q = \angle R = (180^\circ - 20^\circ)/2 = 80^\circ$. The bisector of $\angle Q$ splits it into two 40° parts, so $\angle SQR = 40^\circ$. Forgetting to bisect leaves 80° ; copying $\angle P$ gives 20° ; $(180-20)/\dots$ slips give 70° .

12. (B) The equal sides are AB and BC; the equal angle is at C, which lies between BC and CA — but CA is NOT one of the matched sides, and AB is opposite C. So the matched pair is two sides and a non-included angle: SSA, which does not guarantee congruence. SAS would require the angle between the two equal sides (at B).

13. (A) Composing a reflection in a horizontal axis with a reflection in a perpendicular vertical axis through the same point is exactly a 180° rotation about that point. So order-2 (180°) rotational symmetry is forced. It need not reach order 4 (a non-square rectangle has both mirror lines but only order 2). Order 2 corresponds to 180° , not 90° .

14. (A) $DE = DF$ means the base angles $\angle E$ and $\angle F$ are equal; let each be y . Apex $\angle D = y + 30$. Sum: $(y+30) + y + y = 180$, so $3y = 150$, $y = 50$. Thus $\angle F = 50^\circ$. Using $\angle D = 80^\circ$ as the answer mistakes the apex for a base angle.

15. (C) Corresponding sides: $AB = DE = 13$, $BC = EF = 11$, so $CA = 36 - 13 - 11 = 12$ cm. CA corresponds to FD , but its length comes from the perimeter, not from any angle, so 'cannot be found' is wrong; 13 and 11 are the other two sides.

16. (C) E is symmetric across a horizontal line through its middle bar, but not across a vertical line, and turning it 180° gives a backwards E — no order-2 rotation. H has both a horizontal and a vertical axis (and order-2 rotation). X has both axes and order-2/4 rotation. S has order-2 rotational symmetry but no mirror line.

17. (A) Compute the first hypotenuse: $\sqrt{49 + 576} = \sqrt{625} = 25$ cm. Both now have hypotenuse 25 cm, a leg 7 cm and a right angle — exactly RHS, which uniquely fixes the third leg at 24 cm. The 'other leg unknown' worry is resolved by RHS; this is RHS, not the unreliable SSA, and equal sides give congruence not mere similarity.

18. (C) Parts sum to 12, so each part $= 180^\circ/12 = 15^\circ$. Angles are 45° , 60° , 75° . All differ (scalene) and all are below 90° (acute). The largest is 75° , not 90° or above, so it is neither right-angled nor obtuse; no two are equal, so not isosceles.

19. (D) Base angles of an isosceles triangle are equal: each $= (180^\circ - \angle A)/2 = (180^\circ - 140^\circ)/2 = 40^\circ/2 = 20^\circ$. Forgetting to halve the remaining 40° gives 40° ; mistakenly using $(180-40)/2 = 70^\circ$ treats 140 as an exterior angle it is not.

20. (B) BC and EF are the hypotenuses (opposite the right angle), and AB , DE are legs. So we have a right angle, an equal hypotenuse and an equal leg — precisely RHS. SAS would need the right angle BETWEEN the two given sides, but the hypotenuse is opposite the right angle, so SAS does not apply directly; RHS removes the SSA ambiguity.

21. (D) A regular n -gon has n lines of symmetry and rotational order n . Pentagon: 5 and 5; hexagon: 6 and 6. Differences are $6 - 5 = 1$ for lines and $6 - 5 = 1$ for rotational order.

22. (A) Perpendicular bisector of BC through A means A is equidistant from B and C : $AB = AC$. Perpendicular bisector of AB through C means C is equidistant from A and B : $CA = CB$. Combining: $AB = AC$ and $CA = CB$ give $AB = BC = CA$ — equilateral. One condition alone gives only isosceles; both together force all three equal.

23. (B) The order $ABC \leftrightarrow PQR$ pairs $A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$. Hence $\angle B = \angle Q$ and side $CA \leftrightarrow RP$, so $CA = RP$. The others mismatch the correspondence (e.g. $\angle B$ with $\angle R$, or AB with QR).

24. (D) Let each base angle be b ; apex $= b - 30$. Sum: $(b-30) + b + b = 180$, so $3b = 210$, $b = 70$, apex $= 70 - 30 = 40^\circ$. Reporting the base angle 70° instead of the apex is the trap.

25. (A) An isosceles non-equilateral triangle has exactly one mirror line (through the apex) and no rotational symmetry beyond 360° (order 1). A non-square rectangle has TWO mirror lines and order-2 rotation. A general parallelogram has order-2 rotation but no mirror line. A regular pentagon has 5 mirror lines and order 5.

26. (B) AB and BC are the two sides meeting at vertex B, and $\angle B$ is the angle between them — this is the included angle. Equal AB, equal $\angle B$, equal BC matched the same way is SAS, which guarantees congruence. Because the angle IS included, it is not the ambiguous SSA case.

27. (D) The altitude from the apex bisects BC, so $BM = \sqrt{17^2 - 8^2} = \sqrt{(289 - 64)} = \sqrt{225} = 15$ cm. Then $BC = 2 \times 15 = 30$ cm. Forgetting to double the half-base gives 15; mis-adding gives the others.

28. (C) Reflection is a rigid motion: it preserves all lengths and angles, so area and perimeter are unchanged and the image is congruent (not merely similar). What it does change is orientation — the figure is flipped, reversing left/right handedness. Reflection makes no extra copy and does not enlarge.

29. (B) In DEF, $\angle D + \angle E = 90^\circ$ (since $\angle F = 90^\circ$), and $\angle D = 2\angle E$ gives $3\angle E = 90^\circ$, $\angle E = 30^\circ$, $\angle D = 60^\circ$. Correspondence $ABC \leftrightarrow DEF$ pairs B with E, so $\angle B = \angle E = 30^\circ$. Taking $\angle D = 60^\circ$ (which corresponds to $\angle A$) is the misread.

30. (D) A square coincides with itself after 90° , 180° , 270° and 360° rotations — all four. Its rotational order is 4 (counting 360° as the return), and every multiple of 90° works.

31. (A) Base angles: $\angle B = \angle C = (180^\circ - 100^\circ)/2 = 40^\circ$. The bisector of $\angle B$ makes $\angle DBC = 20^\circ$. In triangle BDC, $\angle BDC = 180^\circ - \angle DBC - \angle C = 180^\circ - 20^\circ - 40^\circ = 120^\circ$. Forgetting the bisector (using $\angle B = 40^\circ$) gives $180 - 40 - 40 = 100^\circ$.

32. (B) (i) is false: an isosceles triangle has a mirror line but only order-1 rotation. (ii) is false: a parallelogram (or the letter S) has order-2 rotation but no mirror line. Both implications fail.

33. (C) BC is opposite $\angle A$ (and EF opposite $\angle D$); the side lies opposite one of the equal angles, not between the two equal angles A and B. That pattern is AAS, not ASA. AAS still guarantees congruence (the third angle is forced, reducing to ASA), so it is congruent — just named correctly as AAS.

34. (B) If 60° maps it onto itself, so do 120° , 180° , 240° , 300° and 360° — six positions, order 6 at least. The order is $360^\circ/60^\circ = 6$, not 3 (which would correspond to 120°) and certainly not 60.

35. (B) With D the midpoint we have $BD = DC$, plus $AB = AC$ and AD common — three equal sides, so SSS. Then $\angle ADB = \angle ADC$ by CPCT; being supplementary on the straight

line BC, each is 90° . RHS would assume the right angle we are trying to prove, and $\angle BAD = \angle CAD$ is not yet established for SAS.

36. (C) Equal corresponding angles give similarity; a 1:1 ratio of one pair of corresponding sides means that pair is equal, which fixes the scale at 1, so all sides are equal — congruent. The named criterion is ASA/AAS, NOT AAA (AAA alone is only similarity).

37. (D) AC is the axis of symmetry of the kite, so reflection across AC fixes A and C and swaps B with D. The segment BD therefore maps onto itself with its endpoints exchanged; since reflection across AC swaps B and D, BD must meet AC at right angles and be bisected by it. It does not become AC or a side.

38. (C) $\angle A = \angle B$ forces $CA = CB$ (sides opposite equal angles), so the triangle is isosceles with apex C. Its axis of symmetry passes through the apex C and the midpoint of the base AB, which is the perpendicular bisector of AB. The line AB or the bisector of BC are not axes unless the triangle is equilateral.

39. (D) Parts sum to 6, so the unit is $180^\circ/6 = 30^\circ$. Angles are 30° , 60° , 90° — a right angle is present, all three differ (scalene). It is not isosceles (no two equal) and not obtuse (largest is exactly 90°).

40. (A) $\angle B$ corresponds to $\angle E$, so $\angle B = x + 30$. With $\angle A = 3x - 10$ and $\angle C = 80^\circ$, angle sum in ABC: $(3x - 10) + (x + 30) + 80 = 180$, so $4x + 100 = 180$, $4x = 80$, $x = 20$. Mis-dropping the constant 100 or using $\angle C = 70$ yields the distractors.

41. (A) A regular n-gon has smallest rotation $360^\circ/n$ and order n. For $n = 8$: $360^\circ/8 = 45^\circ$ and order 8. Pairing 45° with order 4, or halving to 22.5° , mismatches the regular-polygon rule.

42. (B) With $AB = AC$, $\angle BAD = \angle CAD$ (angle bisector) and AD common, triangles ABD and ACD are congruent by SAS. Then $BD = DC$ (so AD bisects BC) and $\angle ADB = \angle ADC$ (CPCT); being supplementary, each is 90° , so $AD \perp BC$. Both properties hold together — the apex bisector is the perpendicular bisector of the base.

43. (B) Reflection preserves all side lengths and angles, so every corresponding part is equal — that is the definition of congruence, which allows flips (orientation reversal). Handedness changes under reflection but does not affect congruence, and this holds for any triangle, scalene or not.

44. (D) With the right angle at A and equal legs $AB = AC = 6$, BC is the hypotenuse: $BC = \sqrt{(6^2 + 6^2)} = \sqrt{72} = 6\sqrt{2}$ cm. Adding legs gives 12 (wrong); 6 ignores the hypotenuse relation.

45. (B) AB and AC meet at A, so $\angle A$ is the included angle between the two given equal sides — this is SAS, which guarantees congruence. ASA needs two angles; only one angle

($\angle A$) is given here, so the ASA label is incorrect even though congruence does hold.

46. (D) Order 6 means the smallest rotation is $360^\circ/6 = 60^\circ$, and every multiple of 60° below 360° works: 60, 120, 180, 240, 300 — five angles. 360° is the identity (not below 360°), and 30° is not a symmetry of an order-6 figure.

47. (A) If the median AD (so $BD = DC$) is also the angle bisector ($\angle BAD = \angle CAD$), the configuration forces $AB = AC$ (the only triangle in which the median from a vertex coincides with the angle bisector is isosceles with apex there). It need not be equilateral or right-angled, and it is certainly not scalene.

48. (A) The parts $4 + 5 + 6 = 15$, so the unit is $45/15 = 3$ cm. $EF = 5$ units = 15 cm. Since BC corresponds to EF ($B \leftrightarrow E, C \leftrightarrow F$), $BC = 15$ cm. $DE = 12$ and $FD = 18$ are the other two sides.

49. (C) Block 'I' looks the same after a 180° turn (order 2) and has both a vertical and a horizontal mirror line. 'S' and 'N' have order-2 rotation but NO mirror line. 'P' has neither rotational symmetry of order 2 nor a mirror line.

50. (D) The apex altitude bisects BC, so the half-base is 4 cm; altitude = $\sqrt{5^2 - 4^2} = \sqrt{9} = 3$ cm. Area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 3 = 12$ cm². Mixing up which is leg vs half-base gives 4; doubling the area mis-applies the $\frac{1}{2}$.

51. (C) A 40° step gives order $360^\circ/40 = 9$. Since the construction explicitly has no reflection, there are 0 lines of symmetry — rotational order does not create mirror lines. 'Order 40' confuses the angle with the order.

52. (D) Triangles ABC and ABD are congruent by SSS ($AC=AD, BC=BD, AB$ common). The whole figure ACBD is symmetric across line AB (it is a kite with axis AB), so reflecting across AB swaps C and D. Hence AB is the perpendicular bisector of CD — it bisects CD at right angles.

53. (C) $DE = DF$ means base angles $\angle E = \angle F$. Let $\angle D = d$; then $\angle E = \angle F = 2d$. Sum: $d + 2d + 2d = 180$, $5d = 180$, $d = 36^\circ$. So $\angle D = 36^\circ$; $\angle E = 72^\circ$ is the base angle, not the apex.

54. (B) A 120° smallest rotation gives order $360^\circ/120 = 3$, and three mirror lines match a regular 3-gon — the equilateral triangle. A regular hexagon has order 6 and six mirror lines; a square has order 4 and four lines. The equilateral triangle's order is 3, not 6.

55. (C) The correspondence $A \leftrightarrow B, B \leftrightarrow C, C \leftrightarrow A$ equates $AB=BC=CA$ (CPCT on sides) and $\angle A = \angle B = \angle C$ (on angles). All three sides equal makes it equilateral. A general triangle does NOT map to itself under a cyclic relabel, so 'nothing' is wrong.

56. (A) Congruent to the 6-8-10 triangle means identical sides 6, 8, 10. Since $6^2 + 8^2 = 36 + 64 = 100 = 10^2$, it is right-angled, so its largest angle (opposite the 10 cm side) is 90° .

Congruence fixes every part, so it can be determined and is not merely acute.

57. (D) A non-square rhombus has its two diagonals as mirror lines (2 lines) and order-2 rotation; a non-square rectangle has the two perpendicular bisectors of its sides as mirror lines (2 lines) and order-2 rotation. Both have exactly 2 lines and order 2 — only the square reaches 4 and order 4.

58. (A) AD bisects the apex angle of an isosceles triangle ($AB = AC$), so it is also the altitude to the base: triangles ABD and ACD are congruent (SAS), making $\angle ADB = \angle ADC$, and being supplementary each is 90° . The base angle 50° is a distractor irrelevant to the right angle at the foot.

59. (A) A rotation is a rigid motion, so all lengths and angles, and hence area and perimeter, are preserved, giving congruence. Unlike reflection, rotation preserves orientation (no flip). Rotation never stretches or resizes, so 'different perimeter' or 'only similar' are wrong.

60. (D) Under reflection, the mirror line is the perpendicular bisector of the segment joining a point and its image. So m crosses PP' at its midpoint F at a right angle, giving $PF = FP' = 10/2 = 5$ cm. m cannot be parallel to PP' (it must cross it), and the two halves are always equal, never 7 and 3.